Surface–like pseudolattices, del Pezzo surfaces, and genus one fibrations

Based on a lecture series by Franco Rota.

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§0.1 About. This course on pseudolattices was a part of a Winter School for students and early career researchers, organised by the UK Algebraic Geometry Network. The lectures were delivered in-person in the University of Warwick, and were transcribed by Parth Shimpi. These notes have undergone several amendments and are not a verbatim recall of the lectures, therefore discretion is advised when using this material. They are available online at `https://pas201.user.srcf.net/documents/2023-ukag-pseudolattices.pdf`. All errors and corrections should be communicated to by email to `parth.shimpi@glasgow.ac.uk`.

Think about homological mirror symmetry on a linear algebraic level.

We recall some of the bilinear–algebraic constructions associated to a smooth projective $\mathbb{C}$–surface S. A *Cartier divisor* D on S. To such D we can associate a line bundle $\mathcal{O}_S(D)$, and this gives a one–to–one correspondence between the set of divisors on S up to rational equivalence and the set of line bundles on S up to isomorphism. Either of these is a group called the *Picard group*.

§0.2 The bilinear pairing on the Picard group. Given C, C' distinct irreducible curves in S and a point $x \in C \cap C'$, one can associate an integer $\mathrm{mult}_x(C \cap C')$ called the *intersection multiplicity* of C, C' at x. If C, C' are locally defined by polynomials f, f' respectively, then the intersection multiplicity is the dimension of $\mathcal{O}_x/(f, f')$. Then the intersection pairing is defined as

$$C \cdot C' = \sum_{x \in C \cap C'} \mathrm{mult}_x(C \cap C')$$

and we extend it linearly to sums of curves.

Fact:$\mathcal{O}_{C\cap C'} = \mathcal{O}_S/\mathcal{O}_S(-C) + \mathcal{O}_S(-C')$, so $C \cdot C' = h^0(S, \mathcal{O}_{C\cap C'})$.

Theorem 0.1. *For* $L, M \in \mathrm{Pic}(S)$,

$$(L \cdot M) = \chi(\mathcal{O}_S) - \chi(L^\vee) - \chi(M^\vee) + \chi(L^\vee \otimes M^\vee)$$

is a symmetric bilinear form that extends $(C \cdot C') = (\mathcal{O}_S(C) \cdot \mathcal{O}_S(C'))$.

Topological interpretation: since S is a complex variety, there is an exponential exact sequence

$$0 \to \mathbb{Z} \to \mathcal{O}_S \xrightarrow{e^{2\pi i(-)}} \mathcal{O}_S^\times \to 0.$$

Then $\mathrm{Pic}(S)$ is isomorphic to $H^1(S, \mathcal{O}_S^\times)$, so taking the long exact sequence in cohomology gives

$$\ldots \to H^0(S, \mathcal{O}_S) \to H^0(S, \mathcal{O}_S^\times) \to H^1(S, \mathbb{Z}) \to H^1(S, \mathcal{O}_S) \to \mathrm{Pic}(S) \xrightarrow{c_1} H^2(S, \mathbb{Z}) \to H^2(S, \mathcal{O}_S) \to \ldots$$

Now the map $H^0(S, \mathcal{O}_S) \to H^0(S, \mathcal{O}_S^\times)$ is simply the exponential map $\mathbb{C} \to \mathbb{C}^\times$ and is in particular surjective. Thus the Picard group sits in an exact sequence

$$0 \to \frac{H^1(S, \mathcal{O}_S)}{H^1(S, \mathbb{Z})} \to \mathrm{Pic}(S) \xrightarrow{c_1} \mathrm{NS}(S) \to 0$$

where $\mathrm{NS}(S) \subset H^2(S, \mathbb{Z})$ is the image of $c_1 : \mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$. We say $\mathrm{NS}(S)$ is the *Néron–Severi group* of S, and the map c_1 is the first *Chern class* map. Given an irreducible curve $C \subset S$, there is a restriction map $H^2(S, \mathbb{Z}) \to H^2(C, \mathbb{Z}) = \mathbb{Z}$ which by Poincaré duality gives $c_1(C) \in H^2(S, \mathbb{Z})$.

Then by construction, the cup product on cohomology gives the intersection product $C \cdot C' = c_1(C) \cap c_1(C')$.

§0.3 Chern characters. If X is a smooth projective variety, then each line bundle $L \in \mathrm{Pic}(X)$ can be assigned a *Chern polynomial* $c(L)_t = 1 + c_1(L)t \in H^\bullet(X, \mathbb{Z})[t]$. We say the *Chern character* is

$$\mathrm{ch}(L) = e^{c_1(L)} = 1 + c_1(L) + \frac{1}{2}c_1(L)^2 + \dots$$

If $E = \bigoplus_{i=1}^r L_i$ is a direct sum of line bundles (or more generally if E is a vector bundle filtered by line bundles $L_1, \dots, L_r$), then the Chern polynomial and character are defined as

$$c(E)_t = \prod_{i=1}^r c(L_i)_t = 1 + c_1(E)_t + c_2(E)t^2 + \dots \mathrm{ch}(E) = \sum_{i=1}^r \mathrm{ch}(L_i) = \sum_{k=0}^{\dim X} \mathrm{ch}_k(E).$$

More generally, the splitting principle lets us assume $c(E)_t$ always has the form $\prod(1 + a_i t)$ for cohomology classes $a_i \in H^2(X, \mathbb{Z})$ called the Chern roots of E. These can be used to define the *Todd class*

$$\mathrm{td}(E) = \prod_{i=1}^r \frac{a_i}{1 - e^{-a_i}}.$$

Exercise 0.2 (Properties of Chern classes). $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$, $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E) \cdot \mathrm{ch}(F)$, $\mathrm{ch}(E^\vee) = \sum_{i=1}^{\dim X} (-1)^i \mathrm{ch}_i(E$

Exercise 0.3 (Chern and Todd classes on surfaces). If E is a vector bundle on a surface S, then

$$\mathrm{ch}(E) = 1 + c_1(E) + \frac{c_1^2(E) - 2c_2(E)}{2}, \qquad \mathrm{td}(E) = 1 + \frac{1}{2}c_1(E) + \frac{1}{12}(c_1^2(E) + c_2(E)).$$

Theorem 0.4 Hirzebruch–Riemann–Roch. *If E is a vector bundle on a smooth projective variety X of dimension n, then*

$$\chi(E) = \sum_{i=0}^n (-1)^i \dim H^i(X, E) = \deg[\mathrm{ch}(E) \cdot \mathrm{td}(T_X)]_n = \int_X \mathrm{ch}(E)\mathrm{td}(T_X)$$

.

Example 0.5. If $X = C$ is a smooth curve and $L = \mathcal{O}_C(\sum n_i p_i)$ is a line bundle, then $\mathrm{td}_C = \mathrm{td}(T_C) = \mathrm{td}(\omega_C^\vee) = 1 + \frac{1}{2}c_1(\omega$ while $\mathrm{ch}(L) = 1 + (\sum n_i)[p]$. Then HRR gives us $\chi(L) = \deg(L) + 1 - g$.

Example 0.6. If $X = C$ is a surface, then $\mathrm{td}_S = (1, -\frac{K_S}{2}, \frac{1}{12}(c_1^2 + c_2))$ and $\mathrm{ch}(L) = (1, L, \frac{L^2}{2})$. Thus HRR gives us

$$\chi(L) = \underbrace{\frac{1}{12}(c_1^2 + c_2)}_{\chi(\mathcal{O}_S)} - \frac{1}{2}(L \cdot K_S) + \frac{L^2}{2} = \chi(\mathcal{O}_S) + \frac{1}{2}(L - K_S) \cdot L.$$

Noether's formula $\chi(\mathcal{O}_S) = \frac{1}{12}(c_1^2 + c_2)$.

Extensions: If E, F are vector bundles, then have Euler pairing $\chi(E, F) = \sum_{i=0}^n (-1)^i \mathrm{ext}^i(E, F)$. Since $\chi(E, F) = \chi(F \otimes E^\vee)$, HRR gives us $\chi(E, F) = \int_X \mathrm{ch}(F)\mathrm{ch}(E^\vee)\mathrm{td}_X$.

Since X is smooth, each sheaf G has a finite resolution by vector bundles $0 \to E_n \to E_{n-1} \to \dots \to E_0 \to G \to 0$. Define $\mathrm{ch}(G) = \sum_{i=0}^n (-1)^i \mathrm{ch}(E_i)$

Define: Grothendieck group of an Abelian category, triangulated category, derived category.

Upshot: $\chi : K_0(\mathcal{C}oh X) \times K_0(\mathcal{C}oh X) \to \mathbb{Z}$ is a bilinear form, not necessarily symmetric.

Define $\ker \chi = \{G \mid \chi(G, H) = 0 \text{ for all } H \in K_0(\mathcal{C}oh X)\}$. In the presence of a Serre functor, this is equal to $\{G \mid \chi(H, G) = 0 \text{ for all } H \in K_0(\mathcal{C}oh X)\}$.

The intersection pairings on $\mathrm{Pic}(X), \mathrm{NS}(X)$ are symmetric. On the other hand, the Euler pairing on $K_0(\mathcal{C}oh X)$ is not usually symmetric.

[Very rough introduction to derived category, cohomology functors, quasi-isomorphisms and roofs, Ext1–Hom correspondence, replacing objects by resolutions, Serre functors]

Theorem 0.7 Serre duality. *Given a smooth projective variety X and complexes $E, F \in \mathbf{D}^b X$, there is a canonical isomorphism*

$$\mathrm{Hom}_{\mathbf{D}^b X}(E, F) \cong \mathrm{Hom}_{\mathbf{D}^b X}(F, E \otimes \omega_X[\dim X])^\vee.$$

References: FM transforms in AG, Gelfand–Manin

Theorem 0.8. *Given a triangulated category $\mathcal{T}$, a Serre functor is an autoequivalence $\mathbb{S} : \mathcal{T} \to \mathcal{T}$ such that* $\mathrm{Hom}(A, B) \cong \mathrm{Hom}(B, \mathbb{S}(A))^\vee$.

Suppose $\mathcal{T}$ is triangulated with a Serre functor $\mathbb{S}$. Then $\ker\chi = \{E \in \mathcal{T} \mid \chi(E, F) = 0 \text{ for all } F \in \mathcal{T}\}$ is equal to $\{E \in \mathcal{T} \mid \chi(F, E) = 0 \text{ for all } F \in \mathcal{T}\}$ since $\chi(E, F) = \chi(\mathbb{S}^{-1}F, E)$. This lets us define the numerical Grothendieck group $K_0^{\mathrm{num}} = K_0/\ker\chi$.

Example 0.9. For E an elliptic curve, $K_0(E)$ is not finitely generated since it contains $\mathrm{Pic}^0(E) \cong E$. On the other hand, $K_0^{\mathrm{num}}(E)$ is isomorphic to $\mathbb{Z}^2$ with basis $[\mathcal{O}_p]$, $[\mathcal{O}_E]$. What's the Euler pairing?

Definition 0.10. A *pseudolattice* is a finitely generated free Abelian group G equipped with a non-degenerate bilinear pairing $\chi : G \times G \to \mathbb{Z}$. We say G is *unimodular* if the induced natural map $\xi : G \to \mathrm{Hom}(G, \mathbb{Z})$ given by $v \mapsto \chi(-, v)$ is an isomorphism.

We say pseudolattices G, G' are isometric if there is a group isomorphism $\varphi : G \to G'$ that respects the pairings.

Definition 0.11. A *Serre operator* on G is an isometry $\mathbb{S} : G \to G$ such that $\chi(u, v) = \chi(v, \mathbb{S}u)$ for all $u, v \in G$.

Exercise 0.12. Defining $\xi^\intercal : G \to \check{G}$ by $v \mapsto \chi(v, -)$, show that a Serre operator exists if and only if ξ is an isometry and in this case $\mathbb{S} = \xi^{-1} \circ \xi^\intercal$.

§0.4 Exceptional bases and mutation. An element $e \in G$ is (numerically) exceptional if $\chi(e, e) = 1$. A (numerical) exceptional sequence is a sequence $(e_1, ..., e_n)$ of exceptional elements such that $\chi(e_i, e_j) = 0$ whenever $j \leqslant i$. An exceptional basis is an exceptional sequence which spans G.

Lemma 0.13. *If there is an exceptional sequence of length* $\mathrm{rank}(G)$ *then the sequence is an exceptional basis and* G *is unimodular.*

Mutations: given $e \in G$ exceptional, we can define operators

$$\mathbb{L}_e(v) = v - \chi(e, v)e, \quad \mathbb{R}_e(v) = v - \chi(v, e)e$$

Given an exceptional sequence $e_\bullet = (e_0, ..., e_n)$ and $0 \leqslant i \leqslant n$, define $\mathbb{L}_i(e_\bullet) = (..., e_{i-1}, \mathbb{L}_{e_i}(e_{i+1}), e_i, e_{i+2}, ...)$ and $\mathbb{R}_i(e_\bullet) = (..., e_{i-1}, e_{i+1}, \mathbb{R}_{e_{i+1}}(e_i), e_i, e_{i+2}, ...)$.

Modelling on $K_0^{\mathrm{num}}(X^2) = H^0(X) \oplus \mathrm{NS}(X)/\mathrm{tors} \oplus H^4(X)$, we make the following definition.

Definition 0.14 (Kuznetsov, Vial). A pseudolattice (G, χ) is *surface-like* if there is a primitive element $p \in G$ such that $\chi(p, p) = 0$, $\chi(p, -) = \chi(-, p)$, and $\chi|_{p^\perp}$ is symmetric.

We say such a p is pointlike. For $u, v \in p^\perp$, write $\chi(u, v) = -u \cdot v$. Write $\mathrm{rk}(v) = \chi(v, p)$, then $p^\perp = \{v \in G \mid \mathrm{rk}(v) = 0\}$.

There is a chain complex $\mathbb{Z} \xrightarrow{p} G \xrightarrow{\mathrm{rk}} \mathbb{Z}$ with middle cohomology $p^\perp/p$ which we call the *Neron-Severi group* $\mathrm{NS}(G)$. This is a finitely generated free Abelian group of rank $\mathrm{rk}G - 2$.

If G has a Serre operator, then the filtration $0 \hookrightarrow \mathbb{Z}p \hookrightarrow p^\perp \hookrightarrow G$ has backward maps $G \xrightarrow{\mathrm{id}-\mathbb{S}} p^\perp \xrightarrow{\mathrm{id}-\mathbb{S}} \mathbb{Z}p \xrightarrow{\mathrm{id}-\mathbb{S}} 0$. In particular $(\mathrm{id} - \mathbb{S})^3 = 0$ i.e. $\mathbb{S}$ is unipotent.

§0.5 The canonical class. If (G, χ) is a surface-like pseudolattice, then there is a unique object $K \in \mathrm{NS}(G) \otimes \mathbb{Q}$ such that for any $v_1, v_2 \in G$, $\chi(v_1, v_2) - \chi(v_2, v_1) = -K \cdot \lambda(v_1, v_2)$ where $\lambda(v_1, v_2) = \mathrm{rk}(v_1)v_2 - \mathrm{rk}(v_2)v_1)$.

In the surface setting, $K = K_X$ is the class of a canonical divisor.

The motivation to consider numerical exceptional collections is to obstruct the existence of exceptional collections in the derived category.

Definition 0.15. If $\mathcal{T}$ is a triangulated category, then $E \in \mathcal{T}$ is exceptional if $\mathrm{Hom}^\bullet(E, E) = \mathbb{C}$. An exceptional collection is a sequence $(E_1, ..., E_n)$ of exceptional objects such that $\mathrm{Hom}(E_i, E_j[k]) = 0$ for all $i > j$, and $k \in \mathbb{Z}$.

We say it is *full* if it generates $\mathcal{T}$.

It is immediate that the classes $[E_1], ..., [E_n]$ form an exceptional basis of K_0^{num}.

The leading conjecture (by Orlov) is that a smooth projective surface over $\mathbb{C}$ admits a full exceptional collection if and only if it is rational.

Other questions:

1. Given an exceptional collection of maximal length $(2 + \mathrm{rk}(\mathrm{NS}\, X))$, when is it full?

2. When do mutations act transitively on the set of all exceptional collections of maximal length?

Answers: for del Pezzo surfaces (i.e. blow ups of $\mathbb{P}^2$ in $\leqslant 8$ points in general position), the answer to both the questions is 'always' (see Pirozhkov '20, Kuleshov–Orlov '94). Also for blow up in 9 points [Krah].

[Cheng–Hayden–Schroll] consider the Fukaya category of a specific Riemann surface to show that the mutation action need not be transitive.

[Krah] found that the Blow up of $\mathbb{P}^2$ in ten points does *not* have a full exceptional collection of maximal length, thus having at least two orbits.

We focus on numerical versions of these.

Theorem 0.16 (Vial). *If X is is a smooth projective surface with $\chi(\mathcal{O}_X) = 1$ i.e. $[\mathcal{O}_X]$ is numerically exceptional, then the following are equivalent.*

1. *X has a numerical exceptional sequence $L_0, ..., L_{n+1}$ of line bundles of maximal length.*
2. *X has a numerical exceptional sequence $E_0, ..., E_{n+1}$ of numerical line bundles (i.e. $rk(E_i) = 1$) of maximal length.*
3. *$K_X^2 = 10 - \rho(X)$ and one of the following holds*
   (a) *$\mathbb{P}^2$–like: $\mathrm{NS}(X) = \langle 1 \rangle$ and $\frac{1}{3}K_X$ is primitive.*
   (b) *$\mathbb{P}^1 \times \mathbb{P}^1$–like: $\mathrm{NS}(X) = U$ (hyperbolic plane) and $\frac{1}{2}K_X$ is primitive.*
   (c) *Blow up of $\mathbb{P}^2$ in $n-1$ like: $\mathrm{NS}(X) = \langle 1 \rangle \oplus \langle -1 \rangle^{n-1}$ and K_X is primitive.*

Main tool [Hille–Perling] is to attach a toric system– given $(E_0, ..., E_{n+1})$, take $D_i = c_1(E_i) - c_1(E_{i-1})$ to get a set $D_1, ..., D_n$ which forms a trigonal unimodular basis for $\mathrm{NS}(X)$. If you assume in addition that the signature is $(1, n-1)$ and there is a canonical class (characteristic element) then such lattices have been completely classified.

Geometric consequences: if X is a smooth projective surface, $\mathcal{O}_X$ is exceptional if and only if $p_g = h^0(K_X)$ and $q = H^1(\mathcal{O}_X)$ both vanish. Such surfaces have been classified by [Kotschik] according to Kodaira dimension.

For $k = -\infty$ these are the minimal surfaces, i.e. $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, and Hirzebruch surfaces $\mathbb{F}_n$ for $n \geqslant 2$.

If $k = 0, 1$ these are the Dolgachev surfaces X_9 (Blow up of $\mathbb{P}^2$ in 9 points) the 9 points form the base locus of a pencil of cubics thus giving an elliptic fibration $X_9 \to \mathbb{P}^1$. You can also apply a log transformation to get $X_9(p_1, ..., p_\ell)$ where you add multiplicity p_i to a smooth fiber F_i. If $k = 0$ this gives the Enriques surface $X_9(2,2)$, all others have $k = 1$.

For $k = 2$ these are some fake projective planes (including Godeaux surfaces).

Theorem 0.17 (Vial). *X smooth projective surface with $p_g = q = 0$. If S is not minimal then there is a numerically exceptional collection of maximal length. If S is minimal then there is a numerically exceptional collection of maximal length except for $X_9(2,2), X_9(2,3), X_9(2,4), X_9(3,3), X_9(2,2,2)$.*

Enriques surfaces therefore do not admit exceptional collections. [Cho] gives an Enriques surface whose blowup has a full exceptional collection.

Kuznetsov recovers a similar classification result– if G is a surface–like pseudolattice of rank $n+2$, we say it is geometric if $\mathrm{NS}(G)$ has signature $(1, n-1)$. We say G is minimal if there are no exceptional objects of rank 0. Then Kuznetsov sets up a 'numerical MMP': if $e \in G$ is an exceptional object of rank 0, then $G_e = {}^{\perp}e$ and have a semi–orthogonal decomposition $G = e \oplus G_e$ with pairing $\begin{pmatrix} 1 & * \\ 0 & \chi_{G_e} \end{pmatrix}$ and $\mathrm{NS}(G) = \mathrm{NS}(G_e) \oplus \langle 1 \rangle$ is an orthogonal decomposition. Then induction and toric systems game recovers the classification.

§1 The Fano/LG correspondence

Start with a Fano variety/orbifold X. The mirror is an LG model $W : Y \to \mathbb{C}$ where Y is a non–compact manifold and W is smooth. Mirrors to surfaces are a source of surface–like pseudolattices– $(K_0^{\mathrm{num}}(\mathrm{FS}(Y, W)), \chi)$ is surfacelike.

But theres more structure– if there is a smooth effective $D \in |-K_X|$ (quasi del Pezzo: X a rational surface with smooth effective anticanonical divisor, which is necessarily an elliptic curve by adjunction) then $i : D \hookrightarrow X$ gives

linear map $i^* : K_0^{num}X \to K_0^{num}D$ induced by a functor $i^* : \mathbf{D}^b X \to \mathbf{D}^b D$ which is spherical. Main reference: [Harder–Thompson]

Definition 1.1 (Addington, Anmo–Logvinenko). A spherical functor is an exact functor $F : \mathcal{A} \to \mathcal{B}$ of triangulated categories with left and right adjoints $L, R : B \to A$ such that the cotwist functor C defined as $C \to \mathrm{id} \to RF$ is an equivalence and $R \cong CL$.

If F is spherical then we can also prove that the twist T defined as $FR \to \mathrm{id} \to T$ is also an equivalence.

Example 1.2. An object $E \in \mathcal{B}$ is n–spherical if $S(E) = E$ and $\mathrm{Hom}(E, E[i]) = \mathbb{C} \oplus \mathbb{C}[n]$. Then the map $\mathbf{D}^b(*) \to \mathcal{B}$ given by $V \mapsto V \otimes E$ is a spherical functor.

Why? Note that $RF = \mathrm{id} \oplus [-n]$ so the Cotwist is $[-n-1]$.

Thus also the twist functor is an equivalence, it is given by the cone of $\mathrm{Hom}(E, X) \otimes E \to X$.

If $D \subset X$ is a smooth anticanonical divisor, then i_* is right adjoint to i^*, and the cotwist is $-\otimes\mathcal{O}(-D) = -\otimes\omega_X$.

[Harder–Thompson] if $f : G \to H$ is a Hom of pseudolattices, f is spherical if there is a right adjoint $r : H \to G$ and $T = \mathrm{id} - fr$, $C = \mathrm{id} - rf$ are invertible.

A spherical functor $f : G \to H$ is relative $(-1)^n$–CY if $C = (-1)^n S$.

If X is a quasi–del Pezzo and D is an elliptic curve, $\mathbb{E} = K_0^{num}(D)$ in the skyscraper–structuresheaf basis $\{a, b\}$ is given by the lattice $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$; this is isomorphic to the lattice $H^1(D, \mathbb{Z})$ with intersection pairing.

We say $f : G \to \mathbb{E}$ is quasi–del Pezzo if f is a relative 0-CY spherical with adjoint r, G is surfacelike with pointlike $p = r(a)$, and G admits an exceptional basis $e_\bullet$ with primitive images in $\mathbb{E}$, and $\mathrm{NS}(G)$ has sign $(1, n-1)$.

Can then show that a is fixed by twist and T acts on $\mathbb{E}$ by $\left(\begin{smallmatrix} 1 & -K_G^2 \\ 0 & 1 \end{smallmatrix}\right)$, further $r(b) \in p^\perp$ and $r(b) = -K_G$ in $\mathrm{NS}(G)$

Notation: given $\mathbb{E} = \langle a, b \rangle$ and $v_i \in \mathbb{E}$, can define a pseudolattice $Z(\{v_i\})$ with formal basis z_i and $\chi(z_i, z_j) = \langle v_i, v_j \rangle_{\mathbb{E}}$.

Classification: up to mutations of $e_\bullet$ and automorphisms of $\mathbb{E}$ of the form $\left(\begin{smallmatrix} 1 & K \\ 0 & 1 \end{smallmatrix}\right)$ any quasiDelpezzo is of the form

$$(A) \quad Z(b, 3a+b, 6a+b, a, ..., a) \to \mathbb{E} \qquad (B) \quad Z(b, 2a+b, 2a+b, 4a+b) \to \mathbb{E}$$

Exercise:what are the surfaces that realise these cases?

Exercise: show that there is an isomorphism $Z(b, 3a+b, 6a+b, a, a) \cong Z(b, 2a+b, 2a+b, 4a+b, a)$

What does this story look like on the mirror? Genus 1 Lefschetz fibrations $W : Y \to \mathbb{C}$ with generic fiber given by a torus and special fibers given by pinched tori. Choose paths from critical points to a point $\infty \in \mathbb{C}$, lift (parallel transport) to get vanishing cycles $Z_i \in F_\infty$. These define cohomology classes $V_i \in H^1(F_\infty, \mathbb{Z}) \cong \mathbb{E}$. Then the vanishing cycles generate $FS(Y, W)$ and so $K_0^{num}(FS(Y, W))$ is isomorphic to $Z(\text{vanishing thimbles } v_i) \cong H_2(Y, F; \mathbb{Z})$. By the Harder–Thompson classification, get 'numerical mirror symmetry statements:$K_0^{num}(X) \to K_0^{num}(D)$ is isomorphic to $K_0^{num}(FS(Y, W)) \to H^1(F, \mathbb{Z})$ and further the isomorphism is given by sequences of mutations. Trick is to take an $e_\bullet$ that is minimal in the sense of $\sum |\mathrm{rk} e_i|^2$.

Question: To prove that $K_0^{num}(FS(Y, W)) \cong Z(\text{vanishing thimbles})$ we use $g = 1$ fibers. Why doesn't this work for higher genus? What if you take fibrations that aren't Lefschetz? What if X is singular (log dP) and $|-K_X| = \emptyset$ but $|-2K_X|$ has sections? McKay correspondence says you should look at an orbifold inside which $\mathbf{D}^b(\tilde{X})$ lives.